ISAC 868 MHz LoRaWAN Antenna-Sensor — System Design & Manufacturing Summary (`Rev E.5`)

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Hardware Revision: Hardware Revision E.5 (`Rev E.5`, `fab/ISAC_868_RevE5_PCBWay_Turnkey.zip`)
Target Radio / MCU: RAKwireless `RAK3172LP-SIP-8-SM-NI` (`STM32WLE5JC` + `SX1262` `RFO_LP`, 3.0 V `DIO3` TCXO `0x06`, EU868)
License: Apache License 2.0

1. Architecture Overview

`ISAC_868_KiCad9` (`Rev E.5`) integrates **868 MHz LoRaWAN Class A telemetry** (+14.0 dBm uplink with adaptive SF7/SF9/SF12 wet-radome link recovery) and **near-field dielectric reflectometry** (+4.0 dBm 3-tone CW probing at 863.1, 866.5, and 869.5 MHz; < 1.25 ms dwell per tone) onto a single 90.00 × 40.00 × 1.60 mm 2-layer FR-4 PCBA (1.51 mm core, 1 oz ENIG finish, 67 populated top-side SMD parts, 0 THT parts). The board mounts inside a flat-roof UV-stabilized ASA/PETG housing (95.7 × 45.7 × 16.1 mm) sealed by a perimeter Parker ORD 5700 EPDM face-seal O-ring and laminated across its thinned 0.80 mm open-East sensing window with 25 µm 3M 467MP acrylic transfer adhesive.

1.1 Shared IFA + Interdigital Evanescent-Field Antenna (`AE1`) & Elliptic Harmonic Filter

- Synthesized with an OpenMP 3D Yee-grid FDTD solver (`rf/fdtd_3d_ant.c` and `rf/ifa_enclosure_mom.py`) incorporating IPC-4552 ENIG (`NiP/Au`) surface resistance (R_s = 23.74 mΩ/□) and substrate/radome dielectric loss. Under unloaded Dry Air inside the enclosure, resonance occurs at f_0 = 869.0 MHz with standalone $|\Gamma_{ant}(866.5\ \text{MHz})|$ = -23.69 dB at `AE1` Pad 1, filter-cascaded $|\Gamma_{in}(866.5\ \text{MHz})|$ = -20.83 dB at `/BR_Q`, worst-case $|\Gamma_{in}|$ ≤ -10.21 dB across 863–870 MHz over FR-4 $\varepsilon_r \in \{4.2, 4.4, 4.6\}$, total efficiency η_{tot} = 45.5% (η_{rad} = 50.0%), and 43.73 MHz/RIU Air-to-Ice resonant sensitivity (869.0 MHz → 834.5 MHz).
- A 3rd-order elliptic low-pass Pi-filter (`C_M1` = 3.6 pF `GJM1555C1H3R6WB01D`, `L_M1` = 8.2 nH `LQW15AN8N2G00D` ‖ `C_MZ` = 0.5 pF `GJM1555C1HR50WB01D`, `C_M2` = 4.3 pF `GJM1555C1H4R3WB01D`) suppresses $2f_0$ (1.733 GHz) by ≥ 37.2 dB and $3f_0$ (2.600 GHz) by ≥ 60.1 dB across dry and water-loaded VSWR states (ETSI EN 300 220-1 limit ≤ -36 dBm).
- A solder-mask-free bare-ENIG opening (`X_gbr = [73.40..87.80] mm, Y_gbr = [6.65..20.80] mm`, with silkscreen excluded from `X_gbr >= 50.0 mm`) keeps LPI mask ridges out of the bonded 0.80 mm sensing window.

1.2 Asymmetric Microstrip Return-Loss Bridge & 2-Point Calibrated Reflectometer

- Built around a Yageo `RT0402BRD078R2L` (8.2 Ω ±0.1%) inline sense resistor `R1`, 0.1% Susumu `RG1005P` reference arms (`R2` = 681 Ω, `R3` = 3.57 kΩ, `R_TERM` = 56 Ω), ±2% Murata `LQW15AN` inductors (`L_Q` = 27 nH, `L_S` = 30 nH), ±0.05 pF `GJM1555C1H` capacitors (`C_S = C_S2` = 1.0 pF), and an Analog Devices `LT5538IDD#TRPBF` logarithmic detector (Z_{IN+} = 291.65 Ω ‖ 1.194 pF at 866.5 MHz).
- Opening a 2.20 × 2.20 mm `F.Cu`/`B.Cu` ground cutout beneath `/BR_S` (C_{par} = 0.1305 pF < 0.15 pF) achieves -0.802 dB main-path insertion loss, 24.81 dB 50 Ω directivity at 866.5 MHz, and 17.60 dB 1st-percentile (`P01`) Air-to-Ice contrast across 2,000 Monte-Carlo trials.
- Probing at +4.0 dBm keeps `LT5538IDD#TRPBF` (`U2`) input power ≤ -25.40 dBm (12.4 dB below its -13 dBm compression point), while unpowered +14 dBm LoRaWAN transmissions induce ≤ 39.5 mV_{pk} at `U2 IN+` up to 6:1 VSWR. Two-point calibration (Dry-Air tare + sintered Al_2O_3 reference offset `cal_ref_cdb`, combined with 0.015 dB/°C residual thermal slope trim and TX-off Listen-Before-Sense blanking) yields 99.72% Water-vs-IPA separation (`P01` margin 0.65 dB ≥ 0.50 dB) in simulation.

1.3 Split-Rail Power Tree & Freestanding C11 Firmware

- `J2` (`B2B-PH-K-S`) and bench headers `J1, J3, J4, J5` are marked `DNP = Yes` for single-pass top SMT assembly; silicone 24–26 AWG battery leads solder from `B.Cu` into `J2` (`Pad 1` = `GND`, `Pad 2` = `+VBAT_RAW`, 4.35 V DC max).
- `Q3` (`AO3401A` ideal-diode P-MOSFET) feeds `U3` (`TPS7A0233DBVR` 25 nA LDO), while `Q1` (`TPS22919DCKR`) powers `LT5538` directly from `VBAT` so `U3` sees at most 24.1 mA (preserving 3.0 V TCXO headroom down to 3.00 V cell

cutoff). `L_SMPS` (`LQH2MCN150K02L` , I_{sat} = 300 mA) feeds the internal `SX1262` buck converter, and `Q2` (`NX3008CBKS,115`) gates battery division on `PA9` for 150 µs.

- Freestanding C11 firmware (15,948 B Flash, 408 B RAM) implements LoRaWAN 1.0.3 EU868 Class A MAC/crypto, §7 Join backoff, rolling 1-hour ETSI duty-cycle accounting, and dual-page CRC-32 Flash NVM with 72-bit `ECCD` `NMI` recovery.

2. Analytical Reflectometer & Ratiometric Signal Model

For an input reflection coefficient $\Gamma_{in}(f_k)$ at `/BR_Q` across probe tones $f_k \in \{863.1, 866.5, 869.5\}$ MHz, the coupled power delivered to `LT5538` (`Port 3`) is governed by bridge directivity leakage S_{31} and reflection tracking $S_{21}S_{32}$:

$$P_{refl}(f_k) = P_{TX} + 20\log_{10}|S_{31}(f_k) + \frac{S_{21}(f_k)S_{32}(f_k)\,\Gamma_{in}(f_k)}{1 - S_{22}(f_k)\,\Gamma_{in}(f_k)}|$$

The logarithmic detector converts $P_{refl}(f_k)$ into baseband voltage $V(f_k) = K_{det}[P_{refl}(f_k) - P_0(T)]$ with nominal slope K_{det} = 19.0 mV/dB and temperature-dependent log-intercept $P_0(T)$. Firmware scales each 16× oversampled ADC reading ratiometrically against the factory-trimmed `STM32WLE5JC` internal bandgap (`VREFINT_CAL` at 3.000 V):

$$V_{DDA,true} = 3000\ \text{mV} \times \frac{16 \times \text{VREFINT_CAL}}{\text{ADC16}_{\text{VREFINT}}}$$

After subtracting the Point-1 Dry-Air baseline tare, firmware forms three features—band-edge slope (`Tilt` , ΔL_{13}), in-band curvature (`Notch` , ΔN_2), and mean coupled power (`Level` , $\bar{L}$):

$$\Delta L_{13} = \frac{V(f_1) - V(f_3)}{19.0\ \text{mV/dB}}$$

$$\Delta N_2 = \frac{½[V(f_1) + V(f_3)] - V(f_2)}{19.0\ \text{mV/dB}}$$

$$\bar{L} = \frac{P_{refl}(f_1) + P_{refl}(f_2) + P_{refl}(f_3)}{3}$$

Because P_{TX} and $P_0(T)$ enter $V(f_1)$, $V(f_2)$, and $V(f_3)$ additively, common-mode PA output variations (ΔP_{TX}) and first-order detector intercept drift ($\Delta P_0(T)$) cancel identically in ΔL_{13} and ΔN_2, leaving only residual slope drift (≈ 0.015 dB/°C) compensated on $\bar{L}$ via `ADC_IN12` die temperature.

3. Summary Specifications & 3D-FDTD Benchmarks

Subsystem Metric	Design / Simulated Value	Target
PCB & Assembly Counts	92 footprints (67 SMD pop, 0 THT; 10 DNP incl. `J2` ; 15 `BOARD_ONLY`), 131 tented vias	0 ERC, 0 DRC, 0 DNP paste
Surface Finish, Mask & CPWG Z_0	IPC-4552 ENIG; 0.035 mm global / 0.025 mm `U1,U2` NSMD (≥ 0.180 mm dams); Z_0 = 51.90 Ω (51.2–57.1 Ω, RL ≥ 26.8 dB)	`ImpedanceControlled = False`
Bridge Through Loss & Directivity	$\|S_{21}\|$ = -0.802 dB, Directivity(866.5 MHz) = 24.81 dB (C_{par} = 0.1305 pF)	≥ 22.0 dB (< 0.15 pF)
2-Point Calibrated 2,000-Trial MC	Air-to-Ice `P01` = 17.60 dB (100% yield); Water-vs-IPA `P01` = 0.65 dB (99.72% yield)	`P01` ≥ 8.0 dB / ≥ 0.50 dB
Enclosed `AE1` Match & Efficiency	f_0 = 869.0 MHz, $\|\Gamma_{ant}\|_{866.5}$ = -23.69 dB (`AE1`), $\|\Gamma_{in}\|_{866.5}$ = -20.83 dB (`/BR_Q`), η_{tot} = 45.5%	$\|\Gamma_{in}\|$ ≤ -10.0 dB all ε_r

Subsystem Metric	Design / Simulated Value	Target
Harmonic Pi-Filter ($2f_0$ / $3f_0$)	37.20 dB (1.733 GHz) / 60.08 dB (2.600 GHz) enclosed (37.46 / 60.65 dB water)	≥ 28 dB / ≥ 45 dB
Weatherproof Enclosure (95.7 × 45.7 × 16.1 mm)	Flat +8.50 mm roof + 0.80 mm open-East ASA/PETG window + 25 µm 3M 467MP + Parker ORD 5700 O-ring (M2.5 × 22 mm corner screws)	Watertight 2-manifold (0 clashes)

3D-FDTD 3-Tone Benchmark Table (+4.0 dBm CW Sensing, h_{sup} = 2.0 mm, 2-Point Calibrated)

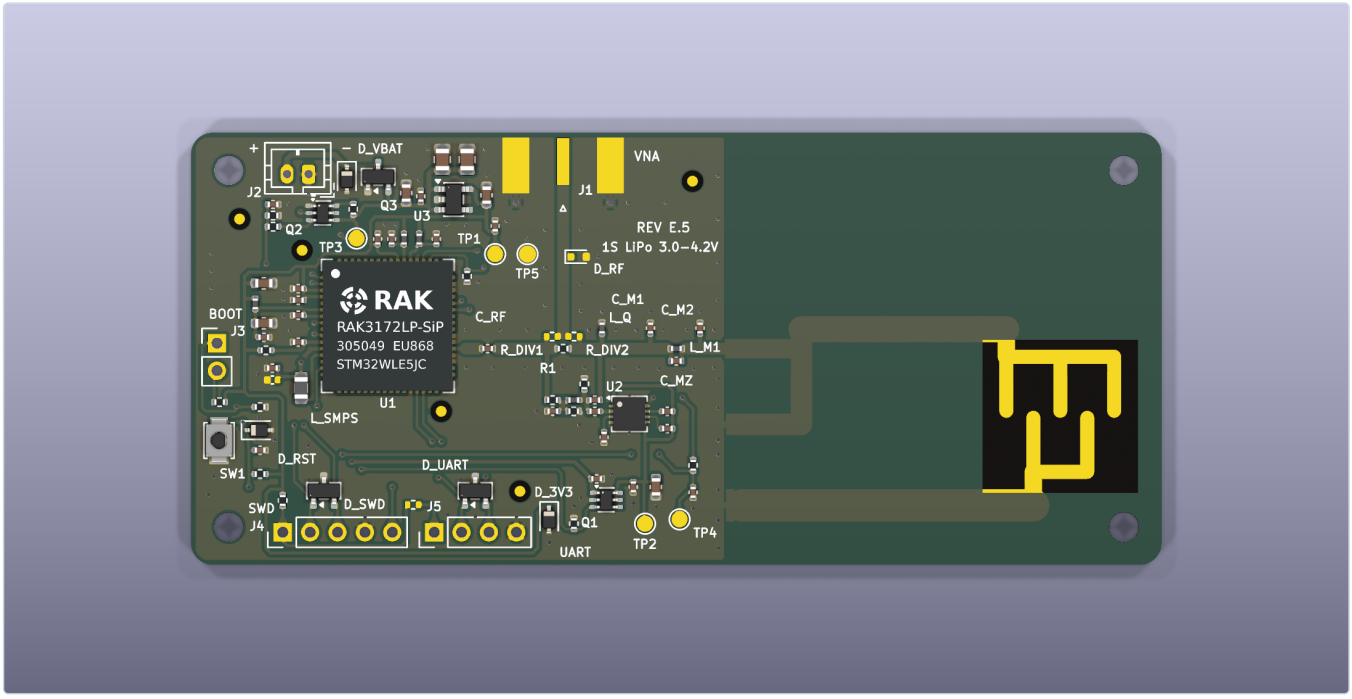
(Reference planes: $|\Gamma_{ant}|$ is standalone at `AE1` Pad 1; $|\Gamma_{in}|$ is cascaded with the elliptic Pi-filter at `/BR_Q`.)

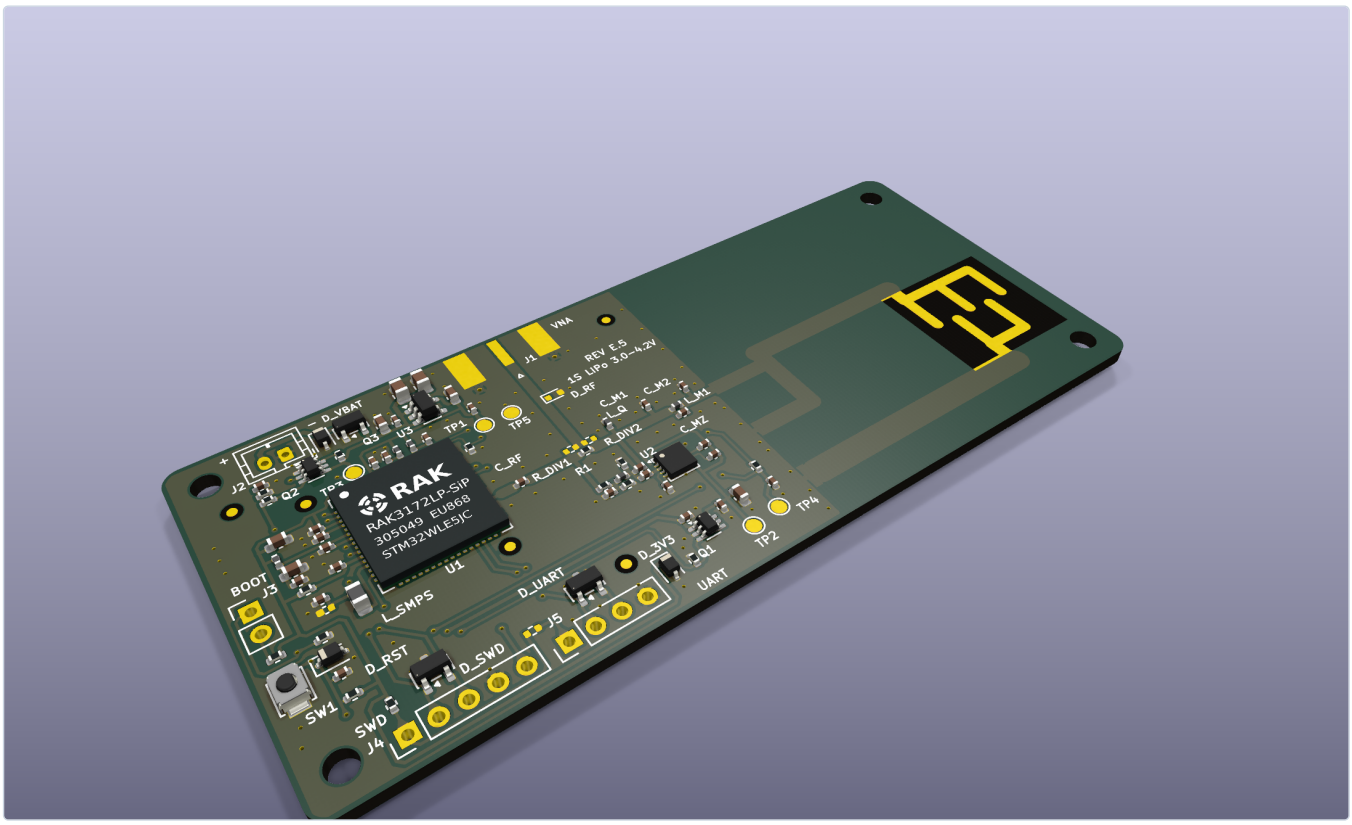
External Superstrate (2.0 mm on 0.80 mm ASA/PETG)	ε_r^* (868 MHz)	f_{res} [MHz]	$\lvert\Gamma_{ant}\rvert$ 866.5 [dB]	Cascaded $\lvert\Gamma_{in}\rvert$ (f_1 / f_2 / f_3) [dB]	Tilt ΔL_{13} [dB]	Notch ΔN_2 [dB]	Level $\bar{L}$ [dBm]	`FPort 10` State
Dry Air	1.00 - j0.000	869.0	-23.69	-18.79 / -20.83 / -18.33	-3.44	+4.88	-45.66	`0 = DRY_AIR`
PTFE / Oil	2.10 - j0.0002	849.0	-7.04	-8.72 / -7.14 / -5.99	-1.23	+0.15	-33.26	`1 = OIL`
Glaze Ice	3.20 - j0.003	834.5	-2.55	-3.65 / -2.89 / -2.32	-0.01	+0.06	-28.63	`2 = GLAZE_ICE`
Alumina (Al_2O_3)	9.80 - j0.002	783.0	-0.36	-0.36 / -0.53 / -0.69	+1.56	+0.03	-26.19	`3 = IPA` (Cal Ref)
IPA (868 MHz)	6.29 - j6.090	794.0	-1.63	-2.27 / -2.19 / -2.12	+1.45	+0.04	-27.75	`3 = IPA`
Water Film	79.9 - j3.810	776.0	-0.46	-0.71 / -0.72 / -0.71	+1.22	+0.04	-26.35	`4 = LIQUID_WATER`

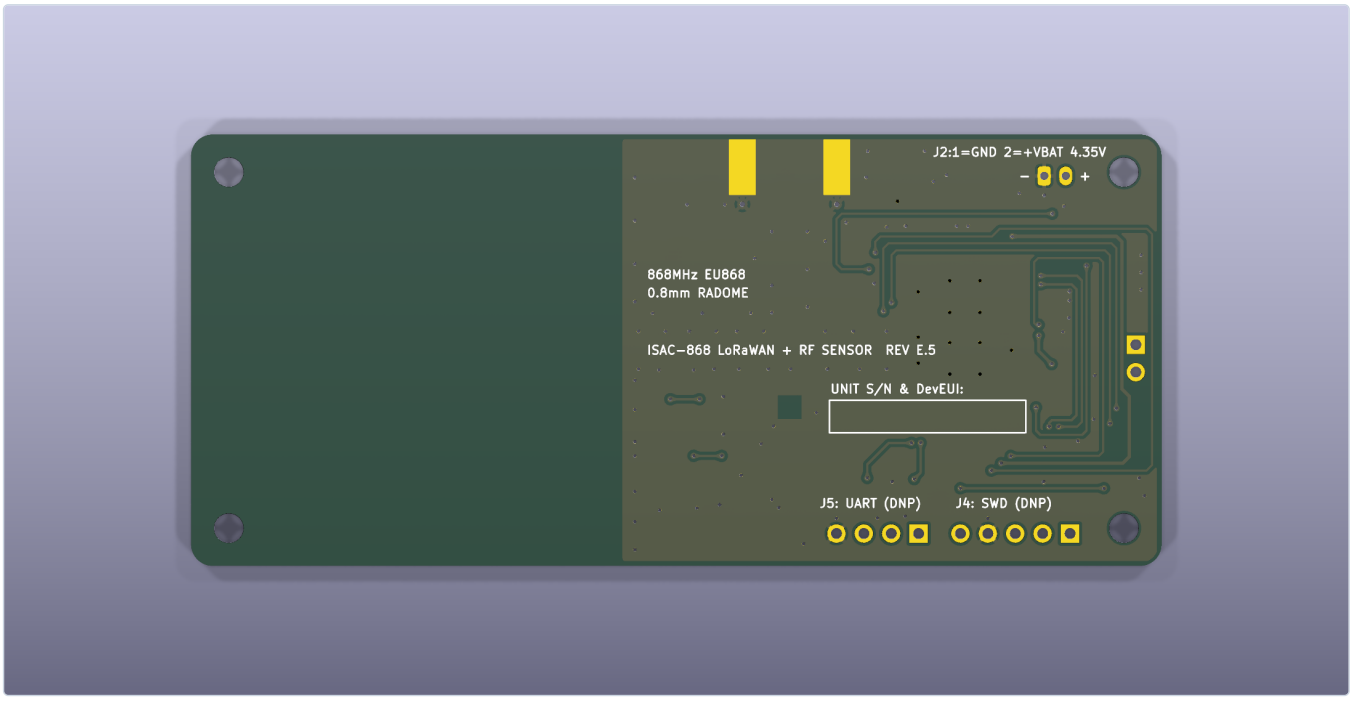
4. Physical Sensing Limitations & Deployment Boundary Conditions

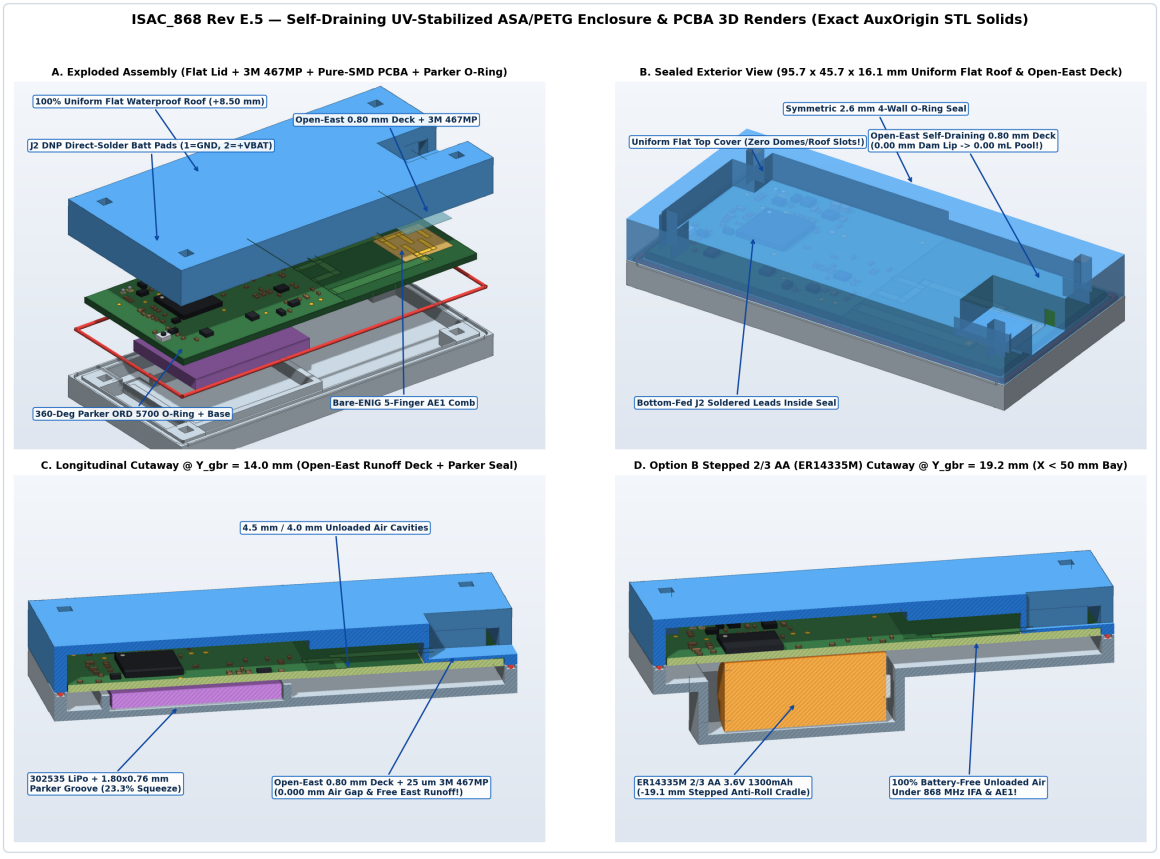
1. **Film-Thickness vs. Permittivity Degeneracy ($\varepsilon_r' \times h_{film}$):** Planar fringing-field sensors measure an effective superstrate capacitance weighted over the evanescent decay length $\delta_E \approx (g + w)/\pi \approx 0.80$ mm [1]. Benchmarks in Section 3 assume a continuous h_{sup} = 2.0 mm layer (> $2\delta_E$). A sub-millimeter water film (h_{sup} < 0.3 mm) or sparse droplets only partially load the evanescent zone and can traverse intermediate (ΔL_{13}, $\bar{L}$) loci similar to thicker oil or ice layers; thermal gating (≤ +4.0 °C for `GLAZE_ICE`) and multi-interval persistence should be applied in field analytics.
2. **Narrowband Skirt Compression & Mounting Counterpoise Coupling:** Because EU868 restricts probing to a 6.4 MHz window (0.74% fractional bandwidth), media with $\varepsilon_r' \geq 6$ pull f_0 below 800 MHz, compressing Water-vs-IPA contrast onto the high-reflection skirt (~0.23 dB in ΔL_{13}, ~1.4 dB in $\bar{L}$). Furthermore, on a 50 × 40 mm ground plane ($0.145\lambda_0$), `B.Cu` chassis currents participate in radiation [2]; mounting directly onto structural steel alters input impedance unless spaced off by a dielectric bracket or tared in-situ via `--cal-dry-mv`.
3. **2-Layer FR-4 Tolerance & Out-of-Band Detector Selectivity:** Uncontrolled 2-layer FR-4 (ε_r = 4.4 ± 0.2) and 0402 pad capacitance around `C_MZ = 0.5 pF` require first-article VNA verification (`0-02`), while the broadband 40 MHz–3.8 GHz `LT5538` input relies on Listen-Before-Sense blanking until an 868 MHz SAW pre-filter is added in `Rev F.0`.

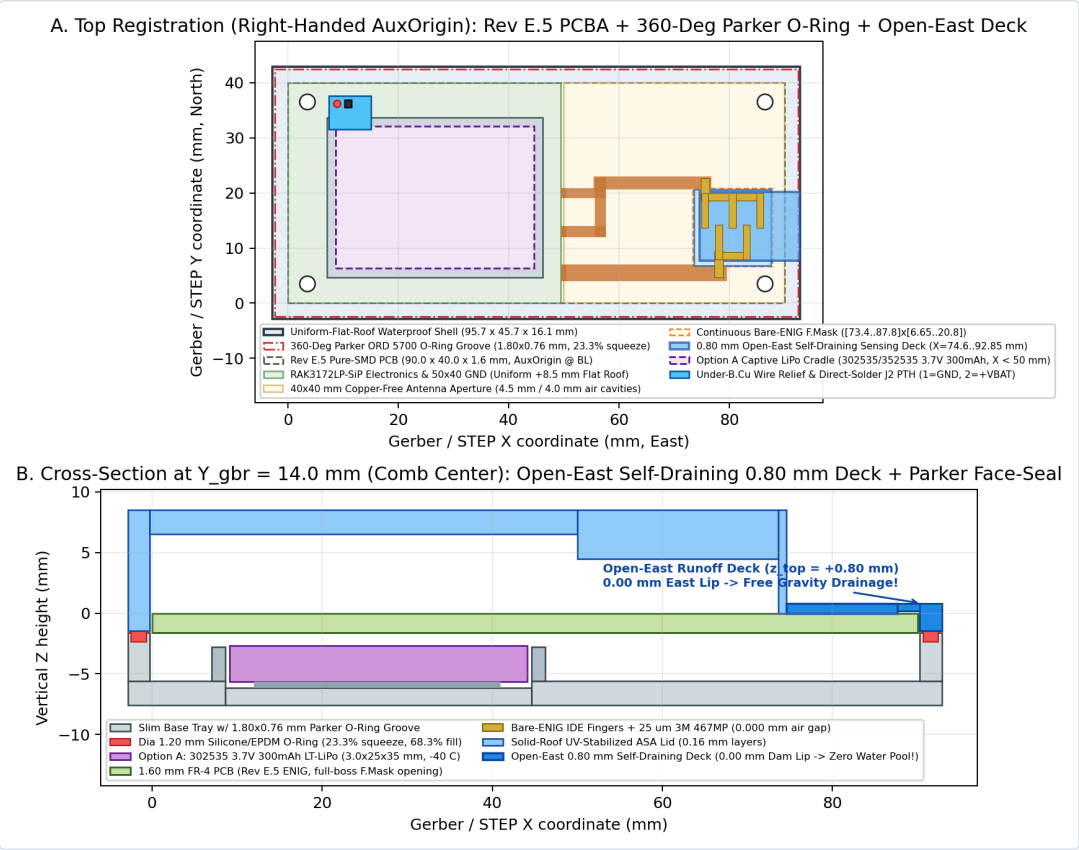
5. CAD & Mechanical Renderings (Rev E.5)






J2:1=GND 2=+VBAT 4.35V
868MHz EU868
0.8mm RADOME
ISAC-868 LoRaWAN + RF SENSOR REV E.5
UNIT S/N & DevEUI:
J5: UART (DNP)
J4: SWD (DNP)


ISAC_868 Rev E.5 — Self-Draining UV-Stabilized ASA/PETG Enclosure & PCBA 3D Renders (Exact AuxOrigin STL Solids)
A. Exploded Assembly (Flat Lid + 3M 467MP + Pure-SMD PCBA + Parker O-Ring)
100% Uniform Flat Waterproof Roof (+8.50 mm)
Open-East 0.80 mm Deck + 3M 467MP
J2 DNP Direct-Solder Batt Pads (1=GND, 2=+VBAT)
360-Deg Parker ORD 5700 O-Ring + Base
Bare-ENIG 5-Finger AE1 Comb
B. Sealed Exterior View (95.7 x 45.7 x 16.1 mm Uniform Flat Roof & Open-East Deck)
Symmetric 2.6 mm 4-Wall O-Ring Seal
Uniform Flat Top Cover (Zero Domes/Roof Slots!)
Open-East Self-Draining 0.80 mm Deck (0.00 mm Dam Lip -> 0.00 mL Pool!)
Bottom-Fed J2 Soldered Leads Inside Seal
C. Longitudinal Cutaway @ Y_gbr = 14.0 mm (Open-East Runoff Deck + Parker Seal)
4.5 mm / 4.0 mm Unloaded Air Cavities
302535 LiPo + 1.80x0.76 mm Parker Groove (23.3% Squeeze)
Open-East 0.80 mm Deck + 25 um 3M 467MP (0.000 mm Air Gap & Free East Runoff!)
D. Option B Stepped 2/3 AA (ER14335M) Cutaway @ Y_gbr = 19.2 mm (X < 50 mm Bay)
ER14335M 2/3 AA 3.6V 1300mAh (-19.1 mm Stepped Anti-Roll Cradle)
100% Battery-Free Unloaded Air Under 868 MHz IFA & AE1!



6. Selected Literature References

1. A. V. Mamishev, K. Sundara-Rajan, F. Yang, Y. Du, and M. Zahn, "Interdigital sensors and transducers," *Proc. IEEE*, vol. 92, no. 5, pp. 808–845, May 2004, doi: `10.1109/JPROC.2004.826603`.
2. G. Marrocco, "The art of UHF RFID antenna design: Impedance-matching and size-reduction techniques," *IEEE Antennas Propag. Mag.*, vol. 50, no. 1, pp. 66–79, Feb. 2008, doi: `10.1109/MAP.2008.4494504`.
3. B. D. Wiltshire, K. Mirshahidi, K. Golovin, and M. H. Zarifi, "Robust and sensitive frost and ice detection via planar microwave resonator sensor," *Sens. Actuators B, Chem.*, vol. 301, art. no. 126881, Dec. 2019, doi: `10.1016/j.snb.2019.126881`.
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5. U. Kaatze, "Complex permittivity of water as a function of frequency and temperature," *J. Chem. Eng. Data*, vol. 34, no. 4, pp. 371–374, Oct. 1989, doi: `10.1021/je00058a001`.
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